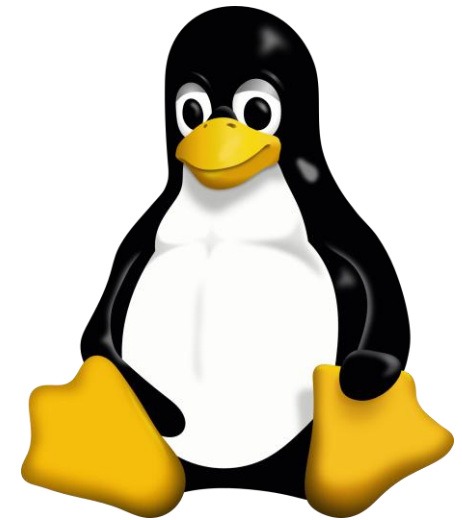


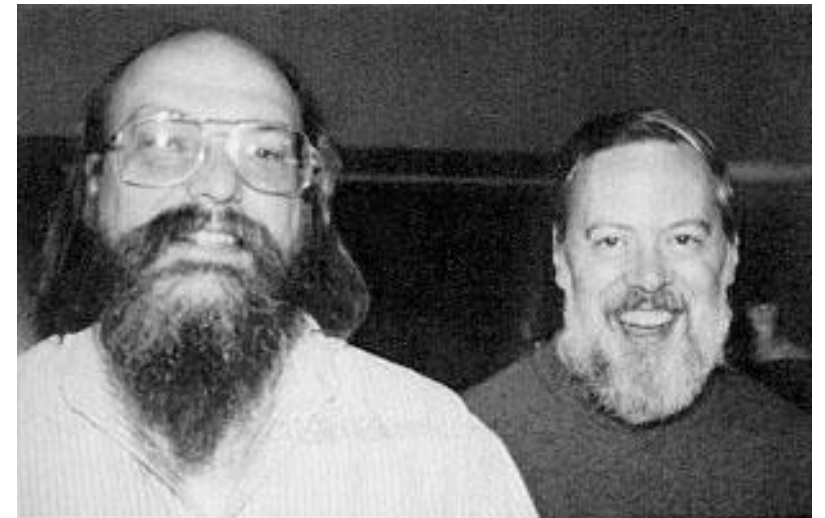
Introduction to Linux and bash

1. History of Linux
2. Why do we use Linux?
3. General Concepts
4. The Shell
5. Definitions



History of Linux

- The History of Linux is defined by several historical developments that came together in 1991 in the first ever Linux operating system
- These steps were:
 - ▶ The development of Unix operating systems by the AT&T Bell Lab
 - ▶ The Rise of operating systems similar to Unix
 - ▶ The GNU Project by Richard Stallman
 - ▶ The Development of a useful kernel by Linus Torvalds



- Originally result of scientific research of the Bell Lab (Dennis Ritchie, Brian Kerningham, Ken Thompson et. al)
- First systems came with the source code
- Written originally in assembly language and later adapted to C
- Named in contrast to MIT's Multics operating system
- Was improved at the Berkeley University by Ken Thompson, Chuck Haley and Bill Joy
- Berkeley software distribution (BSD) derived from this work
- Led to the BSD-family of operating systems in the 90's

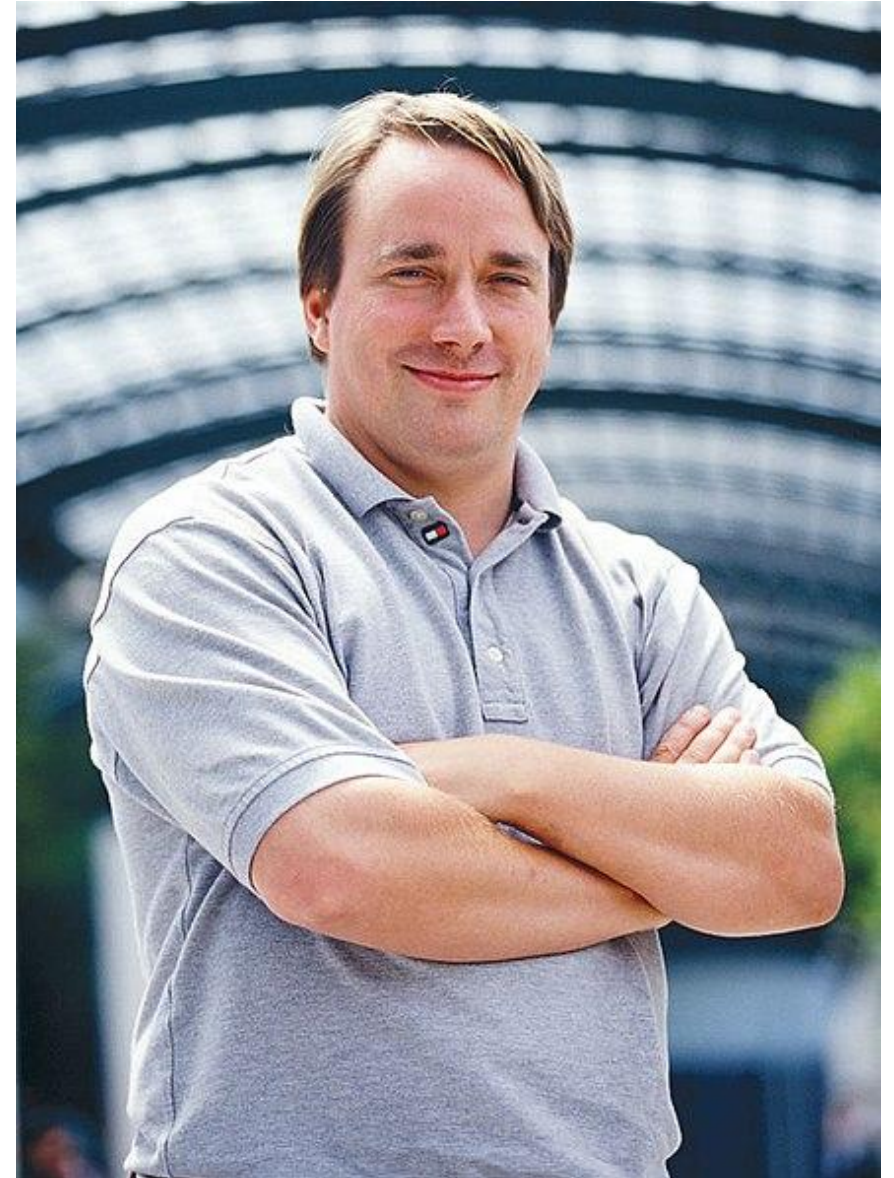
- After the development of Unix, alternative operating systems with similar functionality were created
- On the one hand there were the Onyx systems
- On the other hand there was Sun Microsystems a company that was created by students of the Stanford University
- Both used Unix systems on their commercial available computers and thus marked the beginning of commercialising such operating systems



- Richard Stallman started the GNU project (GNU's Not Unix!) in 1983
- Main aim was to supply the users with an operating system that gives them all liberties [7]
- Open-source versions of the Unix suite of utilities found in BSD
- GNU is also a kind of software license
- All commands in theses courses are derived from GNU



- Linus Torvalds published his first Linux kernel, the core of an operating system, in 1991
- Started as a study of processor architecture in Helsinki
- Has to this day still the authority to decide what gets included into the Linux kernel and what not
- Adopted the GNU license in 1992 and rapidly gathered developers
- Combined the GNU suite of utilities with the new operating system kernel (GNU/Linux)



- Originally Torvalds named his operating System Freax, a mix of the words free, freak and x
- He considered Linux as well but thought it was to egotistical
- Ari Lemmke at Helsinki University of Technology did not like the name and changed it to Linux without asking Torvalds
- Later Torvalds accepted the change
- Originally the name was only intended for the kernel Torvalds developed but it quickly was adopted for the combination of the kernel and any free software under the GNU license and thus the name GNU/Linux was born

Why do we use Linux?

- Shares a functional heritage with Unix systems associated with sciences (Irix, SunOS/Solaris)
- Semi liberal license and source code availability makes it easy for researchers to adjust everything to fit their needs
- Community support and a “storm of enthusiasm” lead to a critical mass of development of new tools
- Wide range of tools available for the user (compiler, scientific libraries, debugger, etc...) that are part of the standard operating system
- **Performance, functionality and portability**

- Many mainframes, large computers with higher computing powers, often use GNU/Linux as operating system
- Aside of it being freely available the construction of the operating system allows for easy multi-processing computing, that means running calculations on more than one processor at the same time



```
1 #!/bin/bash
2 mkdir Linux
3 mkdir Python
4 mkdir Objectorientated
5 mkdir Biopython
6 mkdir Databases
7 mkdir Strings
8 mkdir Alignments
9 mkdir Blast
10 mkdir Structure
11
```

- Since Linux allows us to tackle many different tasks with the help of simple commands, we can automate repetetive tasks with the help of bash scripts
- These scripts are a collection of commands that are executed one after the other and thus produce (hopefully) always the same results
- Especially useful for tasks that you repeat on a weekly, monthly or daily basis

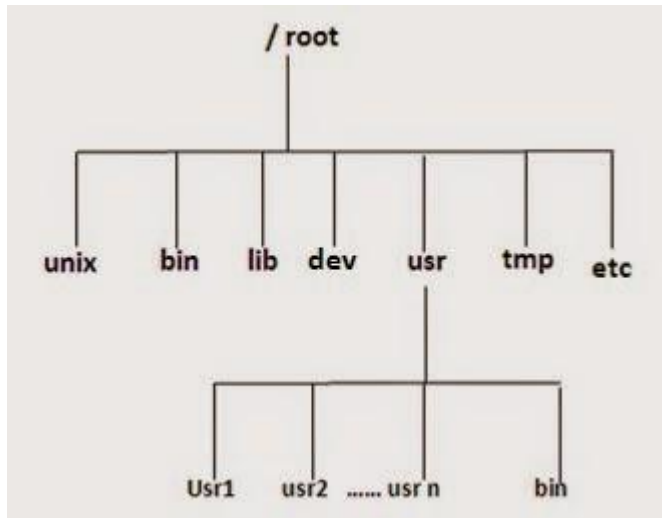
General Concepts



ComputerHope.com

- “Kernel“ of an operating system is the core used by the software to “talk” with the hardware
- Consists of a modular system of drivers to generate a standardized environment
- Works in his own memory the kernel-space
- Responsible for relocating memory and time to system and user processes as well as interacting with files
- Is the basis of an operating system

- Each file and directory has a name
- Names contain letters, numbers and special characters
 - Blanks should be avoided
- Each file has an unique path to its location (e.g. /home/student2/read-write.txt)
- A file name has to be unique within one directory
- Files with the same name can coexist in different directories though
 - We can have a file called “Myfile.txt” in the directory “Sports” as well as in the directory “Texts”
- File names are case-sensitive (“myfile.txt” and “Myfile.txt” can be in the same directory)
- Extensions are ALWAYS part of the filename (file1 is NOT the same as file1.txt)



- Linux filesystems are organized in a tree-like structure
- The top or bottom level is the root
- Unlike in windows there are no drives, drive letters (C:\) or any separate conceptual “space” for storage hardware
- New hardware will come in the form of a “file system” attached (mounted) to some arbitrary point in the directory structure
- When working in a Shell we have to maneuver through this file system using specific commands

The Shell

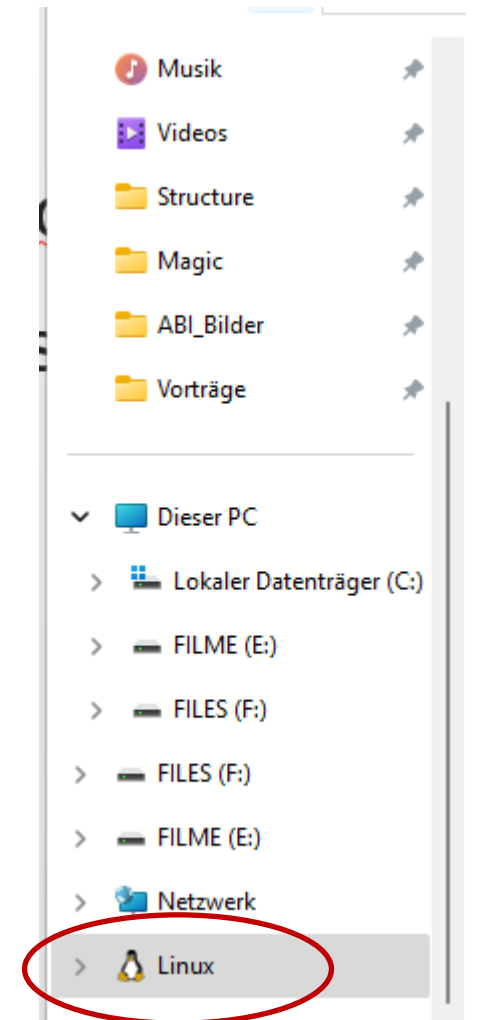
- This is an environment in which you can interact with the kernel in the form of commands
- Defines the syntax of complex command line operations and shell scripts
- We use bash a successor of the vulnerable “Bourne Shell”, called sh
- BASH: “Bourne Again Shell”

```
mars@marsmain ~$ pwd
/home/mars
mars@marsmain ~$ cd /usr/portage/app-shells/bash
mars@marsmain /usr/portage/app-shells/bash$ ls -al
total 130
drwxr-xr-x 3 portage portage 1024 Jul 25 10:06 .
drwxr-xr-x 33 portage portage 1024 Aug 7 22:39 ..
-rw-r--r-- 1 root root 35808 Jul 25 10:06 ChangeLog
-rw-r--r-- 1 root root 27002 Jul 25 10:06 Manifest
-rw-r--r-- 1 portage portage 4645 Mar 23 21:37 bash-3.1_p17.ebuild
-rw-r--r-- 1 portage portage 5977 Mar 23 21:37 bash-3.2_p39.ebuild
-rw-r--r-- 1 portage portage 6151 Apr 5 14:37 bash-3.2_p48-r1.ebuild
-rw-r--r-- 1 portage portage 5988 Mar 23 21:37 bash-3.2_p48.ebuild
-rw-r--r-- 1 portage portage 5643 Apr 5 14:37 bash-4.0_p10-r1.ebuild
-rw-r--r-- 1 portage portage 6230 Apr 5 14:37 bash-4.0_p10.ebuild
-rw-r--r-- 1 portage portage 5648 Apr 14 05:52 bash-4.0_p17-r1.ebuild
-rw-r--r-- 1 portage portage 5532 Apr 8 10:21 bash-4.0_p17.ebuild
-rw-r--r-- 1 portage portage 5660 May 30 03:35 bash-4.0_p24.ebuild
-rw-r--r-- 1 root root 5660 Jul 25 09:43 bash-4.0_p28.ebuild
drwxr-xr-x 2 portage portage 2048 May 30 03:35 files
-rw-r--r-- 1 portage portage 468 Feb 9 04:35 metadata.xml
mars@marsmain /usr/portage/app-shells/bash$ cat metadata.xml
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE pkgmetadata SYSTEM "http://www.gentoo.org/dtd/metadata.dtd">
<pkgmetadata>
  <herd>base-system</herd>
  <use>
    <flag name='bashlogger'>Log ALL commands typed into bash; should ONLY be
      used in restricted environments such as honeypots</flag>
    <flag name='net'>Enable /dev/tcp/host/port redirection</flag>
    <flag name='plugins'>Add support for loading builtins at runtime via
      'enable'</flag>
  </use>
</pkgmetadata>
mars@marsmain /usr/portage/app-shells/bash$ sudo /etc/init.d/bluetooth status
Password:
* status: started
mars@marsmain /usr/portage/app-shells/bash$ ping -q -c1 en.wikipedia.org
PING rr.esams.wikimedia.org (91.198.174.2) 56(84) bytes of data.

--- rr.esams.wikimedia.org ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 2ms
rtt min/avg/max/mdev = 49.820/49.820/49.820/0.000 ms
mars@marsmain /usr/portage/app-shells/bash$ grep -i /dev/sda /etc/fstab | cut --fields=3
/dev/sda1 /boot
/dev/sda2 none
/dev/sda3 /
mars@marsmain /usr/portage/app-shells/bash$ date
Sat Aug 8 02:42:24 MSD 2009
mars@marsmain /usr/portage/app-shells/bash$ lsmod
Module Size Used by
rndis_wlan 23424 0
rndis_host 8696 1 rndis_wlan
cdc_ether 5672 1 rndis_host
usbnet 18688 3 rndis_wlan,rndis_host,cdc_ether
parport_pc 38424 0
fglrx 2388128 20
parport 39648 1 parport_pc
itCO_wdt 12272 0
i2c_i801 9300 0
mars@marsmain /usr/portage/app-shells/bash$
```

- sh – original Unix shell
- **bash – written as a replacement/extension of sh**
- csh – C shell based on the C programming language developed in the late 1970s
- tcsh – enhanced version of C shell
- ksh – Korn shell developed in the early 1980's, backward compatible with sh, but with some features of csh
- zsh – extended version of sh, developed in 1990
- dash – developed as replacement for bash in Debian

- In this course we work with a windows subsystem for Linux
- It allows us to open a Bash shell in an Ubuntu Environment without the need of installing a complete Ubuntu system
- After the Installation is finished a new entry appears at the end of the left side panel of the windows explorer
- From there you are able to access your home directory, the directory where your shell starts, as follows:
 - ▶ Double click on Ubuntu
 - ▶ Double click on home
 - ▶ Double click on the directory with your username



Definitions

- The **Terminal** is the place we work in during this course
- A **Command** is something we can type into our terminal to execute a certain action
- **Options** allow us to adjust the behavior of commands
- We can hand over **Arguments** to commands to work on specific things
- **Directories** are places on your computer that can contain files and other directories, sometimes also called folders
- **Files** can be everything from images, to text files or spreadsheets
- **Current Working Directory** is the directory our terminal is associated with (where we currently are)

- The **Parent Directory** of a directory is the directory which contains the current directory
- All directories in a another directory are the so called **Child Directories** of the corresponding directory

1. https://en.wikipedia.org/wiki/Berkeley_Software_Distribution (Last edited: 22.12.2022, Last visited: 02.01.2023)
2. https://en.wikipedia.org/wiki/History_of_the_Berkeley_Software_Distribution (Last edited: 20.12.2022, Last visited: 02.01.2023)
3. https://en.wikipedia.org/wiki/History_of_Linux (Last edited: 29.3.2023, Last visited: 03.04.2023)
4. https://en.wikipedia.org/wiki/Onyx_Systems (Last edited: 15.3.2023, Last visited: 03.04.2023)
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6. <https://en.wikipedia.org/wiki/GNU> (Last edited: 29.11.2022, Last visited: 02.01.2023)
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8. https://en.wikipedia.org/wiki/Linux_kernel (Last edited: 14.12.2022, Last visited: 02.01.2023)
9. <http://eeasynotes.blogspot.com/2014/11/unix-file-system-unix-file-structure.html> (Last edited: 10.11.2014, Last visited: 02.01.2023)
10. [https://en.wikipedia.org/wiki/Shell_\(computing\)](https://en.wikipedia.org/wiki/Shell_(computing)) (Last edited: 15.12.2022, Last visited: 02.01.2023)